ANUPAM YADAV

Assoicate Professor

Department of Mathematics & Computing
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Research Interests

Optimization, Evolutionary Algorithms, xAI, Large Language Models

Education

2013: **PhD, Mathematics**, *Indian Institute of Technology*, Roorkee.
Optimization, Swarm Intelligence, Evolutionary Algorithms, Soft Computing

2006–2008 : **Master of Science, Mathematics**, *Institute of Science, BHU*, Varanasi.

Teaching & Research Experience

Indian Institute of Technology Roorkee

2019 – 2021: ***Junior Research Fellow, CSIR***.
Advisor : **Dr Kusum Deep**, *Professor HAG (Ret.), Department of Mathematics*, IIT Roorkee

2011 – 2013: ***Senior Research Fellow, CSIR***.
Advisor : **Dr Kusum Deep**, *Professor, Department of Mathematics*, IIT Roorkee

NorthCap University

2013 – 2014: ***Assistant Professor***, *Department of Applied Sciences*.

Korea University, Seoul, Republic of Korea

2014 – 2014: ***Research Professor***, *College of Engineering*.
Advisor : **Dr. Joong Hoon Kim**, *Professor, College of Engineering*, Korea University

National Institute of Technology Uttarakhand

2014 – 2018: ***Assistant Professor***, *Department of Mathematics*.

Dr BR Ambedkar National Institute of Technology Jalandhar

2018 – 2022: ***Assistant Professor***, *Department of Mathematics and Computing*.

2022 – ***Associate Professor***, *Department of Mathematics and Computing*.

Sponsored Research Project

2022 – 2025: **Core Research Grant (MATRICS)**, *Development of Projection Algorithm for Solving Convex Feasibility Problems arise in Quantum Computation. SERB-DST Govt. of India*.

Patent

2026: **System and Method for Coordinated Heterogeneous Unmanned Aerial Vehicle Swarm with Decoy-Based Protection against Radar-Guided Threats**, *Inventors: Shrishti Chamoli, Anupam Yadav*, Status: Filed.

International Collaborations

2023 **Visiting Researcher**, *Korea University, Seoul*.

2024 **Visiting Researcher**, *University of Jaen, Spain*.

2025 **Visiting Researcher**, *Yonsei University, Seoul, South Korea*.

2026 **Visiting Researcher**, *Kyugnpook National University, Daegu, South Korea*.

PhD Supervision

2020: ***Anita***, *Design and Analysis of Optimization Algorithm for High-Order Graph Matching*.

2021: ***Indu Bala***, *A Comprehensive Learning Gravitational Search Algorithm And Its Applications*.

2024: ***Dikshit Chauhan***, *Improvised Artificial Electric Field Algorithms for Optimization Problems*.

Submitted: ***Deepika Khurana***, *Development and Analysis of MOEAS*.

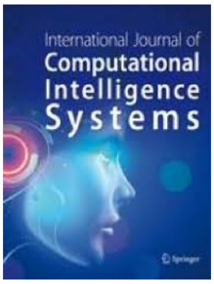
Ongoing: ***Shrishti Chamoli***, *Optimizing Various Functionalities of Drone Swarm using Metaheuristic Algorithms*.

Ongoing: ***Nitin Gupta***, *Explainability of Evolutionary Algorithms*.

Ongoing: ***Ms Vidhi***, *Joined in Jan 2025*.

Ongoing: ***Mr Ram Sundar***, *Joined in Jan 2025*.

Editorial Work (Managing Editor)



Publications

Journal Articles

2027 **[1]** Dikshit Chauhan, Bapi Dutta, Indu Bala, Niki van Stein, Thomas Bäck, and Anupam Yadav. Large language models and evolutionary computation: A critical review of bidirectional interaction, automated algorithm design, and co-adaptive systems. *Computer Science Review*, volume 63, page 101050, 2027.

2026 **[2]** Deepika Khurana and Anupam Yadav. A binary multi-objective optimization algorithm for large-scale feature selection problems. *Evolving Systems*, volume 17, page 5. Springer, 2026.

[3] Deepika Khurana, Awdhesh Kumar Bind, and Anupam Yadav. A multi-objective neural network approach for solving the non-linear fixed charge transportation problem. *Cluster Computing*, volume 29, page 239. Springer, 2026.

[4] Nitin Gupta and Anupam Yadav. An interpretability framework for ant colony optimization in continuous domains. *Swarm and Evolutionary Computation*, volume 107, page 102449. Elsevier, 2026.

[5] Dikshit Chauhan, Deepika Khurana, and Anupam Yadav. Chaotic strategies-enhanced artificial electric field algorithm for combinatorial feature selection. *Journal of Combinatorial Optimization*, volume 51, page 34. Springer, 2026.

[6] Dikshit Chauhan, Nitin Gupta, and Anupam Yadav. An explainable reconfiguration-based artificial electric field algorithm for industrial and reliability-redundancy allocation problems. *Evolving Systems*, volume 17, page 94. Springer Berlin Heidelberg Berlin/Heidelberg, 2026.

[7] Shrishti Chamoli and Anupam Yadav. An ensemble of swarm-based evolutionary learning strategies for uav path-planning problem. *Concurrency and Computation: Practice and Experience*, volume 38, page e70536. Wiley Online Library, 2026.

2025 **[8]** Dikshit Chauhan, Anupam Yadav, and Rammohan Mallipeddi. Eaefa-r: Multiple learning-based ensemble artificial electric field algorithm for global optimization. *Knowledge-Based Systems*, page 113453. Elsevier, 2025.

[9] Dikshit Chauhan, Anupam Yadav, and Sung-Bae Cho. Aefa-fdb: A score-based artificial electric field algorithm for optimal reactive power dispatch problem with renewable and load demand uncertainties. *Applied Soft Computing*, page 113292. Elsevier, 2025.

[10] Dikshit Chauhan, Shivani, Donghwi Jung, and Anupam Yadav. Advancements in multimodal differential evolution: A comprehensive review and future perspectives - artificial intelligence review. *SpringerLink*. Springer Netherlands, Aug 2025.

2024 **[11]** Dikshit Chauhan and Anupam Yadav. Stability and agent dynamics of artificial electric field algorithm. *The Journal of Supercomputing*, volume 80, pages 835–864. Springer US New York, 2024.

[12] Dikshit Chauhan and Anupam Yadav. A comprehensive survey on artificial electric field algorithm: Theories and applications. *Archives of Computational Methods in Engineering*, pages 1–53. Springer Netherlands Dordrecht, 2024.

[13] Dikshit Chauhan and Anupam Yadav. An archive-based self-adaptive artificial electric field algorithm with orthogonal initialization for real-parameter optimization problems. *Applied Soft Computing*, volume 150, page 111109. Elsevier, 2024.

[14] Dikshit Chauhan, Anupam Trivedi, and Anupam Yadav. U-aefa: Online and offline learning-based unified artificial electric field algorithm for real parameter optimization. *Knowledge-Based Systems*, volume 305, page 112636. Elsevier, 2024.

[15] Indu Bala, Anupam Yadav, and Joong Hoon Kim. Optimization for cost-effective design of water distribution networks: a comprehensive learning approach. *Evolutionary Intelligence*, pages 1–33. Springer Berlin Heidelberg, 2024.

2023 **[16]** Deepika Khurana, Anupam Yadav, and Ali Sadollah. A non-dominated sorting based multi-objective neural network algorithm. *MethodsX*, volume 10, page 102152. Elsevier, 2023.

[17] Dikshit Chauhan, Anupam Yadav, and Ferrante Neri. A multi-agent optimization algorithm and its application to training multilayer perceptron models. *Evolving Systems*, pages 1–31. Springer Berlin Heidelberg Berlin/Heidelberg, 2023.

[18] Dikshit Chauhan and Anupam Yadav. Optimizing the parameters of hybrid active power filters through a comprehensive and dynamic multi-swarm gravitational search algorithm. *Engineering Applications of Artificial Intelligence*, volume 123, page 106469. Elsevier, 2023.

[19] Dikshit Chauhan and Anupam Yadav. A crossover-based optimization algorithm for multilevel image segmentation. *Soft Computing*, pages 1–33. Springer, 2023.

[20] Dikshit Chauhan and Anupam Yadav. A competitive and collaborative-based multilevel hierarchical artificial electric field algorithm for global optimization. *Information Sciences*, volume 648, page 119535. Elsevier, 2023.

[21] Dikshit Chauhan and Anupam Yadav. Binary artificial electric field algorithm. *Evolutionary Intelligence*, volume 16, pages 1155–1183. Springer Berlin Heidelberg Berlin/Heidelberg, 2023.

[22] Dikshit Chauhan and Anupam Yadav. An adaptive artificial electric field algorithm for continuous optimization problems. *Expert Systems*, page e13380. Wiley, 2023.

2022 **[23]** Anita Sajwan and Anupam Yadav. A study of exploratory and stability analysis of artificial electric field algorithm. *Applied Intelligence*, volume 52, pages 10805–10828. Springer US New York, 2022.

[24] Indu Bala and Anupam Yadav. Niching comprehensive learning gravitational search algorithm for multimodal optimization problems. *Evolutionary Intelligence*, volume 15, pages 695–721. Springer Berlin Heidelberg, 2022.

2021 **[25]** Anita and Anupam Yadav. An intelligent model for the detection of white blood cells using artificial intelligence. *Computer methods and programs in biomedicine*, volume 199, page 105893. Elsevier, 2021.

2020 **[26]** Anupam Yadav, Ali Sadollah, Neha Yadav, and Joong-Hoon Kim. Self-adaptive global mine blast algorithm for numerical optimization. *Neural Computing and Applications*, volume 32, pages 2423–2444. Springer London, 2020.

[27] Anita, Anupam Yadav, and Nitin Kumar. Artificial electric field algorithm for engineering optimization problems. *Expert Systems with Applications*, volume 149, page 113308. Pergamon, 2020.

[28] Anita and Anupam Yadav. Discrete artificial electric field algorithm for high-order graph matching. *Applied Soft Computing*, volume 92, page 106260. Elsevier, 2020.

2019 **[29]** Anupam Yadav, Neha Yadav, Joong Hoon Kim, and Ali Sadollah. Recent advances in soft computing and its applications. *Journal of Experimental & Theoretical Artificial Intelligence*, volume 31, pages 699–700. Taylor & Francis, 2019.

[30] Hassan Sayyaadi, Ali Sadollah, Anupam Yadav, and Neha Yadav. Stability and iterative convergence of water cycle algorithm for computationally expensive and combinatorial internet shopping optimisation problems. *Journal of Experimental & Theoretical Artificial Intelligence*, volume 31, pages 701–721. Taylor & Francis, 2019.

[31] Indu Bala and Anupam Yadav. Comprehensive learning gravitational search algorithm for global optimization of multimodal functions. *Neural Computing and Applications*, pages 1–36. Springer London, 2019.

[32] Anita and Anupam Yadav. Aefa: Artificial electric field algorithm for global optimization. *Swarm and Evolutionary Computation*, volume 48, pages 93–108. Elsevier, 2019.

2018 **[33]** Ali Sadollah, Hassan Sayyaadi, and Anupam Yadav. A dynamic metaheuristic optimization model inspired by biological nervous systems: Neural network algorithm. *Applied Soft Computing*, volume 71, pages 747–782. Elsevier, 2018.

2017 **[34]** Neha Yadav, Anupam Yadav, Manoj Kumar, and Joong Hoon Kim. An efficient algorithm based on artificial neural networks and particle swarm optimization for solution of nonlinear troesch's problem. *Neural Computing and Applications*, volume 28, pages 171–178. Springer London, 2017.

2016 **[35]** Neha Yadav, Anupam Yadav, and Joong Hoon Kim. Numerical solution of unsteady advection dispersion equation arising in contaminant transport through porous media using neural networks. *Computers & Mathematics with Applications*, volume 72, pages 1021–1030. Pergamon, 2016.

[36] Anupam Yadav, Kusum Deep, Joong Hoon Kim, and Atulya K Nagar. Gravitational swarm optimizer for global optimization. *Swarm and Evolutionary Computation*, volume 31, pages 64–89. Elsevier, 2016.

[37] Anupam Yadav and Kusum Deep. A shrinking hypersphere pso for engineering optimisation problems. *Journal of Experimental & Theoretical Artificial Intelligence*, volume 28, pages 1–33. Taylor & Francis, 2016.

2014 **[38]** Anupam Yadav and Kusum Deep. An efficient co-swarm particle swarm optimization for non-linear constrained optimization. *Journal of Computational Science*, volume 5, pages 258–268. Elsevier, 2014.

2013 **[39]** Anupam Yadav and Kusum Deep. Shrinking hypersphere based trajectory of particles in pso. *Applied Mathematics and Computation*, volume 220, pages 246–267. Elsevier, 2013.

[40] Anupam Yadav and Kusum Deep. Constrained optimization using gravitational search algorithm. *National Academy Science Letters*, volume 36, pages 527–534. Springer India, 2013.

2012 **[41]** Anupam Yadav, Kusum Deep, and Sushil Kumar. An harmonic potential well based particle swarm optimization. *Journal of Information and Operations Management*, volume 3, page 70. Bioinfo Publications, 2012.

[42] Kusum Deep, Anupam Yadav, and Sushil Kumar. Improving local and regional earthquake locations using an advance inversion technique: Particle swarm optimization. *World Journal of Modelling and Simulation*, volume 8, pages 135–141, 2012.

Conference Publications

2025 **[1]** Nitin Gupta, Indu Bala, Bapi Dutta, Luis Martínez, and Anupam Yadav. Enhancing explainability and reliable decision-making in particle swarm optimization through communication topologies. In *Genetic and Evolutionary Computation Conference (GECCO '25 Companion), 10.1145/3712255.3726658*, 2025.

[2] Dikshit Chauhan and Anupam Yadav. Binary aefa for solving combinatorial optimization problems. *Recent Trends in Communication and Intelligent Systems: Proceedings of ICRTCIS 2025*, page 117. Springer Nature, 2025.

2022 **[3]** Dikshit Chauhan and Anupam Yadav. Xor-based binary aefa: theoretical studies and applications. In *2022 IEEE Symposium Series on Computational Intelligence (SSCI)*, pages 1706–1713. IEEE, 2022.

[4] Dikshit Chauhan and Anupam Yadav. Performance of artificial electric field algorithm on 100 digit challenge benchmark problems (cec-2019). *Proceedings of Academia-Industry Consortium for Data Science: AICDS 2020*, page 387. Springer Nature, 2022.

[5] Dikshit Chauhan and Anupam Yadav. A hybrid of artificial electric field algorithm and differential evolution for continuous optimization problems. In *Proceedings of 7th International conference on harmony search, soft computing and applications: ICHSA 2022*, pages 507–520. Springer Nature Singapore Singapore, 2022.

[6] Indu Bala and Anupam Yadav. Optimizing reactive power of ieee-14 bus system using artificial electric field algorithm. In *Congress on Intelligent Systems*, pages 651–665. Springer Nature Singapore Singapore, 2022.

2021 **[7]** Anita, Anupam Yadav, Nitin Kumar, and Joong Hoon Kim. Development of discrete artificial electric field algorithm for quadratic assignment problems. In *Proceedings of 6th International Conference on Harmony Search, Soft Computing and Applications: ICHSA 2020, Istanbul*, pages 411–421. Springer Singapore, 2021.

[8] Anita, Anupam Yadav, and Nitin Kumar. Application of artificial electric field algorithm for economic load dispatch problem. In *Proceedings of the 11th International Conference on Soft Computing and Pattern Recognition (SoCPaR 2019) 11*, pages 71–79. Springer International Publishing, 2021.

2020 **[9]** Harish Sharma, Mukesh Saraswat, Anupam Yadav, Joong Hoon Kim, and Jagdish Chand Bansal. Congress on intelligent systems proceedings of cis 2020, volume 2. In *Congress on Intelligent Systems*, volume 2, page 1, 2020.

[10] Indu Bala and Anupam Yadav. Optimal reactive power dispatch using gravitational search algorithm to solve ieee-14 bus system. In *Communication and Intelligent Systems: Proceedings of ICCIS 2019*, pages 463–473. Springer Singapore, 2020.

[11] Anita, Anupam Yadav, and Nitin Kumar. Artificial electric field algorithm for solving real parameter cec 2017 benchmark problems. In *Soft Computing for Problem Solving 2019: Proceedings of SocProS 2019, Volume 1*, pages 161–169. Springer Singapore, 2020.

2019 **[12]** Indu Bala and Anupam Yadav. Gravitational search algorithm: a state-of-the-art review. *Harmony Search and Nature Inspired Optimization Algorithms: Theory and Applications, ICHSA 2018*, pages 27–37. Springer Singapore, 2019.

2017 **[13]** Neha Yadav, Thi Thuy Ngo, Anupam Yadav, and Joong Hoon Kim. Numerical solution of boundary value problems using artificial neural networks and harmony search. In *Harmony Search Algorithm: Proceedings of the 3rd International Conference on Harmony Search Algorithm (ICHSA 2017) 3*, pages 112–118. Springer Singapore, 2017.

[14] Anupam Yadav, Neha Yadav, and Joong Hoon Kim. A comparative study of exploration ability of harmony search algorithms. In *Harmony Search Algorithm: Proceedings of the 3rd International Conference on Harmony Search Algorithm (ICHSA 2017) 3*, pages 22–31. Springer Singapore, 2017.

2016 **[15]** Neha Yadav, Joong Hoon Kim, and Anupam Yadav. Numerical solution of a class of singular boundary value problems arising in physiology based on neural networks. In *Proceedings of Fifth International Conference on Soft Computing for Problem Solving: SocProS 2015, Volume 2*, pages 673–681. Springer Singapore, 2016.

[16] Anupam Yadav, Neha Yadav, and Joong Hoon Kim. A study of harmony search algorithms: Exploration and convergence ability. In *Harmony Search Algorithm: Proceedings of the 2nd International Conference on Harmony Search Algorithm (ICHSA2015)*, pages 53–62. Springer Berlin Heidelberg, 2016.

2015 **[17]** Neha Yadav, Anupam Yadav, and Kusum Deep. Artificial neural network technique for solution of nonlinear elliptic boundary value problems. In *Proceedings of Fourth International Conference on Soft Computing for Problem Solving: SocProS 2014, Volume 1*, pages 113–121. Springer India, 2015.

[18] Anupam Yadav and Joong Hoon Kim. A niching co-swarm gravitational search algorithm for multi-modal optimization. In *Proceedings of Fourth International Conference on Soft Computing for Problem Solving: SocProS 2014, Volume 1*, pages 599–607. Springer India, 2015.

2014 **[19]** Anupam Yadav and Kusum Deep. A novel co-swarm gravitational search algorithm for constrained optimization. In *Proceedings of the Third International Conference on Soft Computing for Problem Solving: SocProS 2013, Volume 1*, pages 629–640. Springer India, 2014.

2012 **[20]** Anupam Yadav and Kusum Deep. A new disc based particle swarm optimization. In *Proceedings of the International Conference on Soft Computing for Problem Solving (SocProS 2011) December 20-22, 2011: Volume 1*, pages 23–30. Springer India, 2012.

[21] Sushil Kumar, Rama Sushil, Anilesh Kumar, Vijay Kumar Ray, Pratik Ghosh, Sachin Kumar, Swati Shikha, Sourabh Kumar, Ajay Paul, Anupam Yadav, et al. Timely prediction of tsunami using under sea earthquake signals. In *Proceedings of the International Conference on Soft Computing for Problem Solving (SocProS 2011) December 20-22, 2011: Volume 2*, pages 1011–1018. Springer India, 2012.

Edited Books

2024 **[1]** Rajesh Kumar, Om Prakash Verma, Lipo Wang, and Anupam Yadav. Machine intelligence for research and innovations: Proceedings of maitri 2023, volume 2, 2024.

2023 **[2]** Anupam Yadav, Gaurav Gupta, Puneet Rana, and Joong Hoon Kim. Proceedings on international conference on data analytics and computing: Icdac 2022, 2023.

[3] Om Prakash Verma, Lipo Wang, Rajesh Kumar, and Anupam Yadav. Machine intelligence for research and innovations. *Proceedings of MAiTRI*, volume 1, 2023.

[4] Aditya Kumar Singh Pundir, Anupam Yadav, and Swagatam Das. Recent trends in communication and intelligent systems proceedings of icrtcis 2023. *Proceedings of ICRTCIS*, page 1, 2023.

2022 **[5]** Quan Xie, Liang Zhao, Kenli Li, Anupam Yadav, and Lipo Wang. Advances in natural computation, fuzzy systems and knowledge discovery: Proceedings of the icnc-fskd 2021, 2022.

[6] Joong Hoon Kim, Kusum Deep, Zong Woo Geem, Ali Sadollah, and Anupam Yadav. Proceedings of 7th international conference on harmony search, soft computing and applications: Ichsa 2022, 2022.

[7] Gaurav Gupta, Lipo Wang, Anupam Yadav, Puneet Rana, and Zhenyu Wang. Proceedings of academia-industry consortium for data science, 2022.

[8] Mohit Dua, Ankit Kumar Jain, Anupam Yadav, Nitin Kumar, and Patrick Siarry. Proceedings of the international conference on paradigms of communication, computing and data sciences pccds 2021, 2022.

2021 **[9]** Atulya K Nagar, Kusum Deep, Jagdish Chand Bansal, Anupam Yadav, Kapil Ahuja, and Aruna Tiwari. Soft computing for problem solving: Proceedings of socpros 2020, volume 1, 2021.

2020 **[10]** Aruna Tiwari, Kapil Ahuja, Anupam Yadav, Jagdish Chand Bansal, Kusum Deep, and Atulya K Nagar. Soft computing for problem solving, 2020.

[11] Harish Sharma, Mukesh Saraswat, Anupam Yadav, Joong Hoon Kim, and Jagdish Chand Bansal. Congress on intelligent systems, 2020.

[12] Sinan Melih Nigdeli, Joong Hoon Kim, Gebrail Bekdaș, and Anupam Yadav. Proceedings of 6th international conference on harmony search, soft computing and applications ichsa 2020, istanbul. In *Conference proceedings ICHSA*, page 10, 2020.

[13] Joong Hoon Kim, Zong Woo Geem, Donghwi Jung, and Anupam Yadav. Advances in harmony search, soft computing and applications, 2020.

2019 **[14]** Neha Yadav, Anupam Yadav, Jagdish Chand Bansal, Kusum Deep, and Joong Hoon Kim. Harmony search and nature inspired optimization algorithms: Theory and applications, ichsa 2018, 2019.

[15] Anupam Yadav, Satyasai Jagannath Nanda, and Meng-Hiot Lim. Pccda 2023.

[16] Anupam Yadav, Gaurav Gupta, Puneet Rana, and Joong Hoon Kim. Proceedings on international conference on data analytics and computing.

[17] Joong Hoon Kim, Kusum Deep, Zong Woo Geem, Ali Sadollah, and Anupam Yadav. Ichsa 2022.

Book Chapters

2024 **[1]** Anupam Yadav and Shrishti Chamoli. Machine learning applications of evolutionary and metaheuristic algorithms. In *Advanced Machine Learning with Evolutionary and Metaheuristic Techniques*, pages 185–211. Springer Nature Singapore Singapore, 2024.

2019 **[2]** Anupam Yadav, Anita, and Joong Hoon Kim. Convergence of gravitational search algorithm on linear and quadratic functions. *Decision Science in Action: Theory and Applications of Modern Decision Analytic Optimisation*, pages 31–39. Springer Singapore, 2019.

2012 **[3]** Anupam Yadav, Kusum Deep, and Shushil Kumar. Metaheuristic technique for finding earthquake locations in n w himalayan region. In *Advances in Geosciences*, volume 31, pages 1–10. World Scientific, 2012.

Text Book

2015 **[1]** Neha Yadav, Anupam Yadav, and Manoj Kumar. An introduction to neural network methods for differential equations, 2015.

Fellowships & Awards

2015 *Outstanding paper award during Int. Conference at Korea University, Seoul, ROK.*

2014 Selected as a Research Professor under National Research Foundation Korea

2013 Travel Award from NBHM-DAE (Govt. of India) to visit Glasgow, U. K..

2015 Travel Award from CSIR-HRDG (Govt. Of India) to visit Taipei, Taiwan in the year 2011.

2009 Qualified CSIR – JRF (Mathematical Sciences) in the year 2009.

2009 Qualified GATE – 2009 with All India Rank 95.

2009 Awarded Second Prize for the National level paper presentation in MATHSWIZ 2009 at IIT Roorkee.

2008 Qualified JAM-2008.

Other Outreach Activities

2026 Resource Person for the Faculty Development Program on **"Decoding AI and Intelligent Systems Through Mathematical Models"**, Amity University Patna.

2026 Expert Lecture on **"Computational Software (MATLAB & MATHEMATICA)"** during the 3rd Short Term Course, SVNIT Surat.

2026 Expert Lectures on **"Basic Course on Applications of Engineering Mathematics"** as part of a Value Added Course, MIT Art, Design & Technology University, Pune.

2026 Expert Lecture on **"Academic Writing Using LaTeX"** during the 3rd Short Term Course, SVNIT Surat.

2025 Resource Person for the One-Week Faculty Development Program on **"Unlock the Power of AI Metaphors & Computational Intelligence"**, Khalsa College of Engineering and Technology, Amritsar.

2023 Keynote in **4th International Conference on Recent Trends in Communication & Intelligent Systems (ICRTCIS 2023)**, Arya College of Engineering & I.T., Jaipur.

2022 Organizing Secretary of the **International Conference on Data Analytics and Computing (ICDAC-2022)**, Wenzhou-Kean University, China.

2022 General Chair of the **International Conference on Recent Trends in Communication & Intelligent Systems (ICRTCIS 2020)**, Arya College of Engineering & I.T., Jaipur.

2021 Expert Lecture on **"LaTeX"** during the One-Week Short Term Course on **"Academic Writing Using LaTeX"**, SVNIT Surat.

2020 Session Chair for the session on **"Computer Vision"** at the **International Conference on Communication and Intelligent Systems (ICCIS 2020)**.

2020 Session Chair for the session on **"Intelligent System: Algorithms and Applications"** at the **International Conference on Communication and Intelligent Systems (ICCIS 2020)**.

2020 Keynote Session Chair for the talk **"Prospects and Challenges of Deep Learning for Object Detection and Recognition"** at the **Congress on Intelligent Systems (CIS 2020)**.

2020 Session Chair for the session on **"Intelligent Systems and Robotics"** at the **Congress on Intelligent Systems (CIS 2020)**.

2020 Program Committee Member and Reviewer for the **Congress on Intelligent Systems (CIS 2020)**, organized by the Soft Computing Research Society (SCRS).

2020 Organising Secretary of the **Congress on Intelligent Systems (CIS 2020)**, organized by the Soft Computing Research Society (SCRS).

Computer skills

Programming Languages	Python, Matlab, C++
Software Packages	Mathematica

Administrative Responsibilities

2026-	**Head Deparment of Mathematics and Computing**, NIT Jalandhar.
2023-2026	**Associate Dean Examination**, NIT Jalandhar.
2021-	**Coordinator Unnat Bharat Abhiyan**, NIT Jalandhar.
2017-2018	**Centre Incharge CSAB**, NIT Uttarakhand.
2016-2017	**Nodal Officer DASA**, NIT Uttarakhand.
2017-2018	**Associate Dean Academics**, NIT Uttarakhand.
2016-2018	**Warden Boys Hostel**, NIT Uttarakhand.
2015-2016	**Institute Security Incharge**, NIT Uttarakhand.

Subjects Taught

Optimization for Data Science, NIT Jalandhar.
Mathematics-II, NIT Jalandhar.
Linear Algebra, NIT Jalandhar.
Operations Research, NIT Jalandhar.
Numerical Methods, NIT Jalandhar.
Operations Research-II, NIT Jalandhar.
Advanced Optimization Techniques, NIT Jalandhar.
Optimization Techniques, NIT Jalandhar.
Discrete Mathematics, NIT Uttarakhand.